

High-end units may also carry Cybenetics certifications that grade noise (LAMBDA rating) and efficiency (ETA rating). These certifications provide extra assurance that the PSU meets rigorous efficiency and acoustics standards.

Form factors and compatibility

PSUs come in several form factors. ATX remains the standard for mid-tower and full-tower cases; typical dimensions are 150 mm × 86 mm × 140 mm. SFX and SFX-L models are compact units designed for mini-ITX builds. Always check your case's specifications – some small cases only accept SFX units, while very long ATX PSUs (200 mm+) may not fit in compact ATX cases. A new trend in 2025 is the ATX 3.1/PCIe 5.0 specification, which standardizes 12V-2×6 connectors for next-generation GPUs; ensure your chosen PSU includes these connectors or compatible adapters.

Decoding connectors and rails

A PSU's cables connect power to each component. Understanding them helps you plan your build and ensure compatibility:

- **24-pin ATX (20+4):** the main motherboard power cable. It supplies multiple voltages (+3.3 V, +5 V, +12 V and standby) and control signals such as PS_ON# and PWR_OK. It is typically keyed so it cannot be inserted incorrectly.
- **CPU/EPS connectors (4-pin, 4+4 or 8-pin):** provide +12 V directly to the processor's voltage regulator. For boards with 4-pin sockets you can detach one half of an 8-pin connector.
- **PCIe power connectors (6-pin, 6+2-pin, 12VHPWR/12V-2×6):** deliver extra power to graphics cards. High-end GPUs now use 16-pin 12V-2×6 connectors, so ensure your PSU supports them.
- **SATA power (15-pin):** powers SATA SSDs, hard drives and peripherals.
- **Molex (4-pin):** an older connector used for fans and legacy peripherals.

PSUs output power through rails – separate circuits that supply specific voltages. Modern units often combine all +12 V rails into a single high-capacity rail for simplicity and stability. Multiple +12 V rails can provide extra protection by limiting

current on each rail, but they require careful load balancing. Most builders will find single-rail designs easier to work with.

Determining the correct wattage

One of the most critical decisions is choosing the correct wattage. Too little power can cause crashes or system instability, while an oversized unit wastes money and may operate inefficiently. A simple process can help you calculate your power needs.

Step-by-Step: Calculating PSU wattage

1. **List your components and their power requirements.** Check the thermal design power (TDP) of your CPU and GPU, as these are the biggest consumers. A high-end GPU like NVIDIA's GeForce RTX 5080 might have a TDP of ~350 W. A premium CPU such as AMD's Ryzen 9 9900X can draw around 170 W under load. Add RAM, storage devices, fans and LED controllers – typically another 100–150 W.
2. **Account for peak loads.** Components can temporarily draw 20–30 % more power when boosting or starting up. Multiply each TDP by 1.3 to cover these spikes.
3. **Sum the power consumption.** For example, a system with a 350 W GPU, a 170 W CPU and 120 W of peripherals might have a combined draw of $350 \times 1.3 + 170 \times 1.3 + 120 = 455 \text{ W} + 221 \text{ W} + 120 \text{ W} = 796 \text{ W}$.
4. **Divide by efficiency and add headroom.** Because efficiency losses dissipate as heat, you need a PSU that provides more than the system's draw. If you choose an 80 PLUS Gold unit (~90 % efficiency at 50 % load), divide the 796 W load by 0.9 to account for efficiency losses, yielding ~884 W. Finally, add a 20–30 % headroom for future upgrades. This brings the total to roughly $884 \text{ W} \times 1.2 = 1061 \text{ W}$. In this example, a quality 1000 W or 1200 W PSU would be appropriate.
5. **Use a wattage calculator.** Many vendors provide online calculators that automate these steps. Overclockers UK's calculator recommends doubling the system's power draw as a simple rule of thumb, suggesting a 600 W PSU for a 300 W system. Seasonic's calculator automatically adds a 20–30 % margin for peak loads.

Efficiency and 80 PLUS certifications

Energy efficiency affects running costs, heat output and reliability. The 80 PLUS rating system measures efficiency at 20 %, 50 % and 100 % load. A PSU must deliver at least 80 % efficiency at these loads to earn certification. Higher tiers like Gold, Platinum and Titanium indicate progressively better efficiency. The Overclockers UK guide lists minimum efficiencies for each tier: Bronze (82 % at 50 % load), Silver (85 %), Gold (89 %), Platinum (90 %) and Titanium (94 % at 50 % load). Most gaming PCs are well served by Silver or Gold units, balancing cost and performance. A Titanium unit provides maximum efficiency but commands a premium.

